

# satiation

## Satiation and Scarcity after Free AI

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Teaching paper · research study draft · OpenIE [research.openie.dev](https://research.openie.dev)

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**Honesty.** Teaching paper / research study draft, not a final journal PDF. No fabricated citations. Claims keep honesty tags ([UNVERIFIED] / [VENDOR] / [SURROGATE] / [SOFT] as present in the study).  
board\_synth\_claimed=false: software reference stack; surrogate joules are not board watts.

Capability scales. Appetite does not. Design to done.

**Who this is for.** Product leaders, economists adjacent to AI, engineers building agents, and students who need a clear account of why “more intelligence” stops buying human value after a job is done. Pair with the Notational Intelligence teaching paper if you need the commit-gate mechanics.

**Honesty.** Teaching paper and research study draft. Soft historical analogies are marked [SOFT]. Sell-side or press magnitudes are marked [VENDOR] or [PRESS] until primary dashboards are re-checked. Energy numbers from the software reference stay [SURROGATE]. Stack flag:  
board\_synth\_claimed=false. No fabricated citations.

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### What you will learn

- What satiation means as an economic and operational object for work and care, not as a soft preference.
  - Why engagement metrics diverge from economic done, and why that divergence wastes money and attention.
  - How familiar technology cycles move from race to satiation to commodity, with honesty labels on the history.
  - What “free at the margin” means for digital AI chores without claiming free joules or free plant motion.
  - What stays scarce after synthesis gets cheap: physics, liability, and commit. How Open Interface Engineering (OpenIE) meters and Wise Computer Automation (WCA) refuse connect to that scarcity.
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## Glossary

| Term                                                          | Plain meaning in this paper                                                                                                           |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Open Interface Engineering (OpenIE)</b>                    | Energy-aware computing company. Motto in practice: measure, route, eliminate excessive waste.                                         |
| <b>Satiation</b>                                              | Past a completeness threshold, more synthesis or tooling does not increase human-valued completeness for that chore.                  |
| <b>Bliss point</b>                                            | Classical consumer-theory bundle where more of a good reduces utility. Kinship with satiation; cite Andersen and textbook treatments. |
| <b>Economic done</b>                                          | Human-valued finish line for a work or care loop. Orthogonal to model capability ceiling.                                             |
| <b>Work completeness</b>                                      | Ticket closed, patch shipped, report filed, ledger reconciled, acceptance met.                                                        |
| <b>Care completeness</b>                                      | Assessment finished, meds reconciled, family notified, child safe, handoff done.                                                      |
| <b>Free-at-margin</b>                                         | Digital chore synthesis whose marginal price tends toward utility, bundling, or free, while joules and commit remain non-free.        |
| <b>Wise Computer Automation (WCA)</b>                         | Commit nomenclature and gate discipline at the AI-to-machine boundary.                                                                |
| <b>Energy-First Architecture (EFA)</b>                        | Treat energy as part of permission and metering, not as a late marketing slide.                                                       |
| <b>Proposal / Certificate / RefuseReason / CommitDecision</b> | Wire objects of commit law. See the Notational Intelligence paper for full teaching.                                                  |
| <b>Look-Up Table (LUT)</b>                                    | Discrete allow table used in the software reference gate.                                                                             |
| <b>Control Barrier Function (CBF)</b>                         | Optional safe-set certificate conjunct.                                                                                               |
| <b>Model Context Protocol (MCP)</b>                           | Tool transport standard. Not a done detector.                                                                                         |
| <b>System One</b>                                             | Typed decision models that collapse generation tax for known options.                                                                 |
| <b>Artificial General Intelligence (AGI)</b>                  | Broad capability aspiration. Not a pricing excuse for satiated chores.                                                                |
|                                                               | Real meter on real hardware under a stated workload.                                                                                  |

| Term                                                                 | Plain meaning in this paper                        |
|----------------------------------------------------------------------|----------------------------------------------------|
| Device Under Test (DUT)                                              |                                                    |
| Joule tier A / B / C                                                 | Surrogate / post-synthesis estimate / board meter. |
| [SOFT] /<br>[SURROGATE] /<br>[VENDOR] /<br>[PRESS] /<br>[UNVERIFIED] | Honesty tags.                                      |

## Problem in plain language

Most AI forecasts assume infinite appetite.

More capability. More tokens. More seats. More agents. Charts go up and to the right because modelers treat demand for intelligence like demand for oil: elastic, unbounded, forever hungry.

That is the wrong physics for work and care.

Humans satiate. When a job is done (the invoice filed, the child picked up, the patient assessed, the pipeline green) extra intelligence does not create extra value. It creates noise, cost, and overkill. Capability can scale. Appetite cannot.

Unbounded-demand models miss the finish line. They price closed loops as if they were still open markets. They sell Artificial General Intelligence blue-ocean premiums for work that is already at human satiation: low-value chores wearing a frontier costume.

If your strategy assumes infinite appetite for intelligence-as-a-service, you are optimizing a world that does not exist for a large share of digital chores.

## Core thesis

For work and care, satiation is a real stop condition. Digital AI chores are already on a free-at-margin arc for many synthesis tasks. Physics and commit do not discount like software. Open Interface Engineering makes joules visible. Wise Computer Automation makes refuse first-class for both economic done and physical unsafe.

Deck line:

Capability scales. Appetite does not. Design to done.

Product one-liner:

Sell completion. Meter joules. Refuse after the finish line, and refuse before the plant moves without a certificate.

## Definitions and objects

### Satiation

**Name.** Satiation.

**Definition (economic kinship).** The more of a good one already has, the less one will give up for more. Marginal utility falls. It can reach zero or turn negative. Andersen (2001) treats satiation as a structural feature of preference dynamics, not merely taste. Textbook bliss points make the same idea geometric: past the bliss bundle, more reduces utility.

**Definition (operational, this paper).** A task or care loop is satiated when additional synthesis, tooling, tokens, or “AI features” do not increase human-valued completeness (work closed / care closed). Past that threshold, more AI is noise plus cost plus attention tax.

**Why it matters.** Products that assume “more is always better” will overbuild seats, over-prompt agents, and misread engagement as value.

**Example.** You do not want an infinite spell-check. You want the document correct. You do not want unbounded navigation. You want to arrive. You do not want forever-more chat about the same chore. You want the chore closed.

**Scope.** This is narrower than “all goods eventually satiate.” Status goods, military races, and open-ended research can remain non-satiating. The claim targets work and care completeness.

### Economic done versus engagement

| Lens         | Engagement / unbounded model          | Satiation / completeness model                                  |
|--------------|---------------------------------------|-----------------------------------------------------------------|
| Demand       | Always more intelligence useful       | Flat or negative after completeness                             |
| Product KPI  | Tokens, seats, sessions, time-on-task | Time-to-done; refuse rate after done; joules after done         |
| Pricing      | Scarcity premium forever              | Commodity after satiation; premium where physics or commit bind |
| Failure mode | Under-build capacity                  | Overkill after closed loops                                     |
|              | Elastic attention assumed             | Finite attention, trust, hours, responsibility                  |

| Lens             | Engagement / unbounded model      | Satiation / completeness model         |
|------------------|-----------------------------------|----------------------------------------|
| Human constraint |                                   |                                        |
| Safety story     | Soft confidence / more monitoring | Typed refuse + hold at commit boundary |

Attention-economy critiques (Center for Humane Technology; Tim Wu's *The Attention Merchants*) describe engagement maximization as extracting attention rather than serving goals. Regenerating after done is the enterprise cousin of infinite scroll. Same pathology, different costume.

## Free-at-margin

**Name.** Free-at-margin.

**Definition.** For pure information chores with a clear completeness condition, marginal price collapses toward bundling, utility pricing, or “just there,” like spell-check around a text box.

**Why it matters.** Teams confuse temporary synthesis scarcity with durable matter scarcity. Soft moats get eaten by rivals, open weights, cascades, retrieval, and typed decision heads.

**Critical honesty.** Free-at-margin is not free physics. Joules still meter. Plant motion still requires commit. Never write “free AI” as if Landauer and actuators vanished.

## Completeness objects

- **Work completeness:** close the ticket, ship the patch, file the report, reconcile the ledger.
- **Care completeness:** finish the assessment, reconcile meds, notify family, keep the child safe.
- **Technical capacity:** what the system can still generate after done.

Completeness is a stop condition, not a quality vibes score. A done detector that fires too early is under-care. One that never fires is engagement cosplay.

## How it works

### The cycle pattern (race to satiation to commodity)

We have watched related movies. New tech races. People overpay for status and scarcity. Usefulness saturates. The product becomes infrastructure: cheap, expected, boring.

| Technology | Race phase               | Satiation signal  | Commodity endgame        | Honesty                                    |
|------------|--------------------------|-------------------|--------------------------|--------------------------------------------|
| Bicycles   | Novelty, clubs, prestige | “I can get there” | Utility / recreation kit | Boom-bust finance strong (Quinn & Turner); |

| Technology        | Race phase                          | Satiation signal               | Commodity endgame                  | Honesty                                                                 |
|-------------------|-------------------------------------|--------------------------------|------------------------------------|-------------------------------------------------------------------------|
|                   |                                     |                                |                                    | completeness<br>reading<br>[SOFT]                                       |
| Cars              | Horsepower and brand theater        | Reliable trips                 | Mass mobility, used markets        | Pattern<br>[SOFT]                                                       |
| Planes            | Heroic aviation, speed records      | Arrive safely on a schedule    | Seats and fuel math                | Seat commodity strong; heroics-to-schedule<br>[SOFT]                    |
| TVs               | Screen and resolution arms race     | Enough fidelity for content    | Commodity panel + subscription     | Panel commodity strong; “enough fidelity”<br>[SOFT]                     |
| Cell phones       | Spec sheets and cameras             | Always-reachable communication | Pocket utility computer            | Pew smartphone ownership ~91% (2024-2025) strong saturation             |
| AI digital chores | Frontier seats, agents, AGI theater | Chore closed without theater   | Bundled / free-at-margin synthesis | Cost collapse strong (Epoch); chore satiation is the operational thesis |

**Rule.** If your AI wedge is still “more capability” after the user’s finish line, you are selling race-phase theater into a commodity market.

### The free-AI arc, taught as steps

1. **Spell-check was a product.** Then a feature. Then expected air around the text box [SOFT] industry observation for the last stage.
2. **Search and maps** moved from scarce discovery toward expected infrastructure via ads and bundling [SOFT] pattern.

3. **Inference price for fixed performance collapsed.** Epoch AI (Mar 2025 data insight) reports extreme year-scale declines across benchmarks. Epoch AI (Sep 2026, *The plunging price of thought*) reports roughly 47% cost decline per quarter for given performance since about 2023 (about 13x per year). Read primary pages before quoting finer splits.
4. **Open weights and typed decisions** further cannibalize synthesis scarcity. System One-class tools collapse generation tax for known option sets.
5. **Frontier competence slides down the device stack** via distillation, quantization, and table-lookup methods. Treat “~6 months frontier distill” as a planning heuristic only ([**UNVERIFIED**] as a universal law).

Teaching translation: the destination for many digital chores is spell-check economics, not forever scarcity premiums.

### **Care and work: design to done**

Work completeness example: the pull request is merged and tests are green. Extra agent churn is latency and distraction.

Care completeness example: the handoff is done and the person is safe. More generative chat is not more care. Care is not a funnel.

Baumol's cost-disease tradition explains why labor-intensive care resists pure productivity theater. Magnitudes are contested across papers. Consensus enough for teaching: care remains labor-intensive, trust-intensive, and liability-intensive. Artificial intelligence that speeds documentation without a done or refuse discipline can raise measured activity while missing completeness.

Acemoglu and Restrepo (2019) frame automation as task displacement plus possible new-task reinstatement. Completeness-oriented AI is closer to closing a task than inventing engagement. Unbounded agent loops without done detectors look like “so-so automation”: costly activity, weak productivity.

**Product rule.** Ship a done detector before shipping a louder model.

### **What still stays scarce**

If digital intelligence races toward free at the margin, what does not discount?

**Physics does not discount.** Mass, heat, latency of matter, batteries, actuators, factories, clinic floors. You can backfill software onto weaker chips. You cannot wish joules out of Landauer's ledger or torque out of a plant.

**Commit stays scarce.** The hard part is not proposing another action. The hard path is allowing the machine to move, or allowing an irreversible tool to run, when the world pays for the mistake. Soft confidence strings are not certificates. Probability routes. Certificates commit.

**Liability and trust stay scarce.** Someone still owns the signature, the handoff, and the harm.

## Bridge to Wise Computer Automation

In the OpenIE software reference, compose looks like:

```
commit = lut_allow AND energy_ok [AND cbf_ok]
```

On refuse: typed reasons, plant hold, executor not called.

Teaching bridge sentence:

Refuse = economic done U physical unsafe

Two faces of one discipline. You do not pay frontier joules for a chore that is already complete. You do not spend plant joules on an action the gate will not certify.

OpenIE meters the lane in joules (or honest watt-hours), not tokens, seats, or vibe-points as primary scarcity. Measure. Route. Eliminate excessive waste. When the meter is honest, satiation becomes visible: you can see where appetite ended and overkill began.

Landauer honesty for messaging:

- Landauer is a lower bound.
- It does not mean today's large-model inference is near (kT).
- It does mean "information is free" is false in physics.
- Do not slap Landauer on a marketing slide as product nearness.

Horowitz (ISSCC 2014) teaching point: computing is increasingly energy-limited; memory and data movement dominate. Quote carefully. Vendor tokens-per-joule claims stay **[VENDOR]** until Device Under Test metering exists. This stack remains board\_synth\_claimed=false.

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### Worked example: a closed ticket that will not stay closed

A support team uses an agent to draft replies.

Completeness test: customer issue resolved, reply sent, ticket status Done.

Without satiation discipline, the agent keeps "improving" the thread: alternate drafts, summary of the summary, suggested follow-ups the customer did not ask for. Token meters look healthy. Human attention does not. The finish line was already crossed.

With satiation discipline:

1. Done detector fires when acceptance criteria are met.
2. Further synthesis is refused as policy or budget refuse (economic done).

3. If an irreversible action appears (charge a card, delete a record), Wise Computer Automation still requires a Certificate.
4. Meters show pre-done joules separately from post-done joules once instrumentation exists.

Teaching point: refuse after done is not hostility to users. It is respect for finite human days.

### Worked example: care handoff

A care coordinator uses a language model to draft a handoff note.

Completeness test: critical fields filled, medication list reconciled, receiving party confirmed, person in a safe state.

Engagement metrics would reward a long session. Care metrics should reward time-to-safe-state and handoff integrity. Extra generative flourish after the handoff is signed is not care. It is noise beside liability.

### Anti-patterns card

| Anti-pattern                                                        | Why it fails                            |
|---------------------------------------------------------------------|-----------------------------------------|
| Artificial General Intelligence blue-ocean price on satiated chores | Appetite already flat                   |
| Soft confidence as plant permission                                 | Probability is not a certificate        |
| Token price as the only scarcity meter                              | Misses joules and irreversible commit   |
| Infinite agent loops as productivity                                | No completeness stop                    |
| “Free AI” erasing energy or plant cost                              | Physics does not discount               |
| Using rebound to deny satiation                                     | Rebound needs residual elasticity       |
| Landauer cosplay (“we are near $kT$ ”)                              | Orders of magnitude gap; surrogate risk |

### How to teach the claim in under a minute

Say this out loud:

“People finish work and care loops. Models do not know that unless we define done. Cheap synthesis is racing toward free at the margin for digital chores. Joules and irreversible commits do not become free. So meter energy, detect done, and refuse both overkill and unsafe motion.”

If a listener can repeat that without your slides, the teaching landed.

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## Evidence (allowed claims only)

### Theory and literature evidence

- Satiation and bliss-point lineage: Andersen (2001) and standard consumer theory. Theory claim with cites, not a measured agent evaluation.
- Engagement versus done: attention-economy literature (Wu; Humane Technology). Framing plus literature, not a randomized trial.
- Free-at-margin cost curves: Epoch AI inference price work (2025 insight; 2026 plunging-price report). Free-at-margin is not free physics.
- Task automation framing: Acemoglu and Restrepo (2019); Autor task tradition. Contested magnitudes. Frame, do not “solve.”
- Care cost-disease framing: Baumol (1967) and later health-econ debates. Contested magnitudes. Care stays trust and liability intensive.
- Sell-side skepticism existence proof: Goldman Sachs *Gen AI: Too Much Spend, Too Little Benefit?* overview. Treat dollar magnitudes as **[VENDOR/SELL-SIDE]** and re-check the primary PDF before quoting.

### Historical cycle evidence

- Bicycle mania finance: Quinn and Turner (2019) on British cycle shares boom and bust. Strong on race finance. Completeness reading **[SOFT]**.
- Smartphone saturation: Pew Research mobile fact sheet, U.S. smartphone ownership about 91% in 2024 and 2025. Strong saturation evidence.
- Airline seats as yield-managed commodities after deregulation: strong on commodity endgame of seats.

### Design claims (not empirics yet)

Decision rules from the satiation blueprint remain design claims:

1. Design to done; sell completion.
2. Assume soft synthesis moats get eaten.
3. Plan for competence sliding down the device stack.
4. Compete on meter plus commit, not chat theater.
5. Never claim free for energy or plant motion.
6. Ship done detector before louder model.
7. Probability routes; certificates commit.
8. Split meters for pre-done joules versus post-done joules (needed experiment).
9. Segment total addressable market: satiating chores versus non-satiating races.
10. Treat engagement KPIs as hostile in care and work products unless proven aligned.

## **Software-reference kinship**

The commit stack that teaches refuse-before-execute is the same Wise Computer Automation reference used in the Notational Intelligence paper. Seed-1 surrogate episode numbers and Model Context Protocol demo behaviors may be cited only with **[SURROGATE]** and demo labels. They do not prove economic satiation empirically. They prove that refuse can be implemented as typed runtime law.

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## **Common confusions and counters**

### **“Cheaper AI will always unlock infinite demand (Jevons).”**

Rebound requires residual demand elasticity. At a bliss point for a closed chore, direct rebound on that chore collapses toward zero. You do not spell-check forever once the document is correct. Rebound can still appear as new tasks or attention spillover. Product rule: meter post-done joules separately from pre-done joules. Denying global rebound while asserting per-chore satiation without split meters is not allowed as settled empirics.

### **“Status and creative markets never satiate.”**

Often true for status trophies and some creative novelty. False for invoice filing. Segment markets. Do not price satiated chores as Artificial General Intelligence blue ocean.

### **“Military and adversarial races have no bliss point.”**

True enough that the satiation thesis does not apply cleanly there. Commit and safety gates still bind. Scope the claim.

### **“Open-ended research always wants more.”**

True for frontier labs and many science tools. That does not rescue total-addressable-market math for satiated software-as-a-service chores.

### **“Humans do not know when they are done.”**

Sometimes preferences are constructed. Care and liability domains still need stop conditions. “Maybe they want more chat” is a weak excuse for uncapped agents beside a patient handoff.

### **“Engagement is the product.”**

Then say so honestly. You are not in the work or care completeness business. Different ethics, different meters.

## “AI will solve Baumol and end care cost disease.”

Forbidden overclaim. Care stays trust-intensive and liability-intensive. Artificial intelligence can raise activity while missing completeness.

## “Free AI means free energy.”

Hard refuse. Free-at-margin is a price story for digital synthesis. Physics still meters joules. Plants still need certificates.

## “Soft confidence is enough permission.”

Probability routes. Certificates commit. Shared anti-pattern with the Notational Intelligence paper.

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## Limits

1. **No shipped split meter yet** for pre-done versus post-done joules on real workloads. Design rule, not present empiric.
  2. **No done-detector evaluation rates** published here (false-done / false-continue). Future work.
  3. **Care handoff integrity metrics** not measured in this archive.
  4. **Total-addressable-market segmentation study** remains strategy until studied.
  5. **Energy claims on the software reference** are [SURROGATE] only.  
board\_synth\_claimed=false.
  6. **Historical analogies** are partly [SOFT]. Do not upgrade soft pattern recognition into natural law.
  7. **Epoch magnitudes** should be re-read from primary pages before executive citation of exact multiples.
  8. **Goldman Sachs dollar figures** need primary PDF verification before quotation.
  9. **Military and arms-race domains** are out of clean satiation scope.
  10. **No claim that OpenIE has eliminated waste in production fleets.** The lane is meter plus commit discipline.
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## What to do next

1. Pick one chore you ship. Write its completeness test in one sentence.
2. Ask whether your KPI rewards done or rewards lingering.
3. If you meter tokens or seats as primary scarcity, add a joule (or honest watt-hour) view.
4. Separate pre-done spend from post-done spend, even if the split starts as a spreadsheet.
5. Put irreversible actions behind a certificate gate. Read the companion paper: <https://research.openie.dev/papers/ni/>
6. Open figures: <https://research.openie.dev/living/satiation/>
7. Download PDF: <https://research.openie.dev/pdfs/satiation.pdf>

Five piece assembly without mythology:

1. Human satiation bounds demand for work and care. Completeness is the feature.
2. Tech cycles race, then satiate, then commodity. Digital AI chores included.
3. Free at the margin is the software destiny of spell-check-shaped tasks.
4. Physics and commit remain scarce.
5. OpenIE makes the energy interface visible. Wise Computer Automation makes the AI-to-machine commit boundary named, auditable, and refuse-first.

Capability can keep racing. Human appetite will not. Companies that win the commodity phase will design for done: cheapest-correct paths, joule meters, and commit gates that know when to hold.

That is not Artificial General Intelligence theater. That is present-tense engineering for present humans.

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## References

### Economics and theory

- Andersen, E. S. (2001). Satiation in an Evolutionary Model of Structural Dynamics. *Journal of Evolutionary Economics*. <https://doi.org/10.1007/PL00003852>
- Baumol, W. J. (1967). Macroeconomics of Unbalanced Growth. *American Economic Review* 57: 415-426.
- Acemoglu, D. & Restrepo, P. (2019). Automation and New Tasks. *Journal of Economic Perspectives* 33(2). <https://www.aeaweb.org/articles?id=10.1257/jep.33.2.3>
- Gillingham, Rapson & Wagner. The Rebound Effect and Energy Efficiency Policy. <https://gwagner.com/wp-content/uploads/Gillingham-Rapson-Wagner-2015-Rebound-Effect.pdf>
- Christensen, C. *The Innovator's Dilemma* (1997). Christensen Institute overview: <https://www.christenseninstitute.org/theory/disruptive-innovation/>

### Attention and forecasts

- Center for Humane Technology, attention economy overview. <https://www.humanetech.com/youth/the-attention-economy>
- Wu, T. *The Attention Merchants*. <https://www.penguinrandomhouse.com/books/234876/the-attention-merchants-by-tim-wu/>
- Goldman Sachs Research overview: *Gen AI: Too Much Spend, Too Little Benefit?* <https://www.goldmansachs.com/insights/top-of-mind/gen-ai-too-much-spend-too-little-benefit>

### Cost curves and saturation evidence

- Epoch AI. LLM inference price trends (Mar 2025). <https://epoch.ai/data-insights/llm-inference-price-trends>

- Epoch AI. *The plunging price of thought* (Sep 2026). <https://epoch.ai/publications/the-plunging-price-of-thought>
- Pew Research Center mobile fact sheet. <https://www.pewresearch.org/internet/fact-sheet/mobile/>
- Quinn & Turner (2019). British Bicycle Mania, *Cambridge Journal of Economics*.

## Energy honesty and commit kinship

- Landauer, R. (1961). Irreversibility and Heat Generation in the Computing Process.
- Horowitz, M. Computing's Energy Problem (ISSCC 2014).
- Companion teaching paper: Notational Intelligence as Commit Law. <https://research.openie.dev/papers/ni/>
- Anthropic Model Context Protocol. <https://www.anthropic.com/news/model-context-protocol>
- Laya System One. <https://laya-ai.com/system-one-models>

## Local claim discipline

Frozen Allowed / Forbidden / Needs-experiment rows: OpenIE claim ledger for Satiation (research archive). Public teaching cut must not promote Forbidden rows.

## Appendix A. Decision rules card

| ID  | Rule                                                                   |
|-----|------------------------------------------------------------------------|
| R1  | Design to done; sell completion.                                       |
| R2  | Assume soft synthesis moats get eaten.                                 |
| R3  | Plan for competence sliding down devices.                              |
| R4  | Compete on meter + commit.                                             |
| R5  | Never claim free energy or free plant motion.                          |
| R6  | Split pre-done J vs post-done J.                                       |
| R7  | Segment satiating chores vs non-satiating races.                       |
| R8  | Treat engagement KPIs as hostile in care/work unless aligned.          |
| R9  | Care metrics: handoff integrity / safe-state time, not session length. |
| R10 | State joule tier when citing Landauer or Horowitz in product claims.   |

David Charlot is founder of Open Interface Engineering ([openie.dev](https://openie.dev)). Companion teaching paper: *Notational Intelligence as Commit Law*. Style lock: `EDUCATIONAL_PROSE.md`.